DISCRETE MATHEMATICAL STRUCTURES

Student Notes

Modules 1–4

Mathematical Logic
Sets, Relations and Functions
Counting Techniques
Abstract Algebra and Lattices

Artificial Intelligence and Machine Learning
Diploma Programme

Exam-Oriented Notes

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MODULE 1 – MATHEMATICAL LOGIC

The syllabus covers propositional logic, connectives, truth tables, implication, tautology, contradiction, contingency, equivalence, converse, inverse, contrapositive, duality, predicate logic, WFF, quantifiers, natural deduction, rules of inference, methods of proof and resolution.

1. Propositional Logic

Important

High Frequency – 2024 & 2025

Definition

A **proposition** is a declarative statement that has exactly one truth value:

- True (T), or
- False (F).

Examples

- "$2 + 3 = 5$." → Proposition → True
- "10 is an odd number." → Proposition → False

Not Propositions

- "Close the door." → Command
- "What is your name?" → Question
- "How beautiful!" → Exclamation
- "$x + 2 = 5$" → Not a proposition until the value of x is specified.

2. Logical Connectives

Important

High Frequency – 2024 & 2025

Let p and q be propositions.

Connective	Symbol	Meaning
Negation	$\neg p$	NOT p
Conjunction	$p \wedge q$	p AND q
Disjunction	$p \vee q$	p OR q
Implication	$p \rightarrow q$	If p, then q
Biconditional	$p \leftrightarrow q$	p if and only if q

Memory Tip

$\boxed{\wedge = \text{AND}}$

$\boxed{\vee = \text{OR}}$

$\boxed{\neg = \text{NOT}}$

3. Basic Truth Tables

3.1. Negation

$$\neg p$$

p	$\neg p$
T	F
F	T

3.2. Conjunction

$$p \wedge q$$

Conjunction is true **only when both propositions are true**.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Memory Tip

AND needs ALL true.

3.3. Disjunction

$$p \vee q$$

Disjunction is false **only when both propositions are false**.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Memory Tip

OR needs only ONE true.

4. Implication

Important

High Frequency – 2024 & 2025

$$p \rightarrow q$$

means:

If p, then q.

The important point is:

$$\boxed{p \rightarrow q \text{ is false only when } p = T,\ q = F}$$

p	q	$p \rightarrow q$
T	T	T
T	F	**F**
F	T	T
F	F	T

Memory Tip

Implication fails only when TRUE leads to FALSE.

5. "Only If"

This is frequently confusing.

Statement:

"I pass only if you pass."

Let

$$p = \text{I pass}$$

and

$$q = \text{you pass.}$$

Then:

$$\boxed{p \to q}$$

Memory Tip

$$\boxed{\text{A only if B} = A \to B}$$

6. Biconditional

$$p \leftrightarrow q$$

means:

p if and only if q.

It is true when both propositions have the **same truth value**.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

7. Tautology, Contradiction and Contingency

Important

High Frequency – 2024 & 2025

7.1. Tautology

A compound proposition that is **always true**.

Example:

$$p \vee \neg p$$

Therefore:

$$\boxed{p \vee \neg p \text{ is a tautology}}$$

7.2. Contradiction

A compound proposition that is **always false**.

Example:

$$p \wedge \neg p$$

Therefore:

$$\boxed{p \wedge \neg p \text{ is a contradiction}}$$

7.3. Contingency

A proposition that is sometimes true and sometimes false.

Example:

$$p \wedge q$$

has both T and F values.

Therefore it is a contingency.

Quick Comparison

Type	Truth Values
Tautology	All T
Contradiction	All F
Contingency	Both T and F

Memory Tip

T-C-C

- Tautology → Totally true
- Contradiction → Completely false
- Contingency → Changes

8. Logical Equivalence

Two propositions P and Q are logically equivalent if they have the same truth value in every possible case.

Written as:

$$\boxed{P \equiv Q}$$

Important laws:

$$\boxed{\neg(\neg p) \equiv p}$$

$$\boxed{p \to q \equiv \neg p \vee q}$$

$$\boxed{\neg(p \wedge q) \equiv \neg p \vee \neg q}$$

$$\boxed{\neg(p \vee q) \equiv \neg p \wedge \neg q}$$

9. De Morgan's Laws

Very important for simplification.

$$\boxed{\neg(p \wedge q) \equiv \neg p \vee \neg q}$$

$$\boxed{\neg(p \vee q) \equiv \neg p \wedge \neg q}$$

Memory Tip

Break the bracket and change the operator.
AND becomes OR.
OR becomes AND.

10. Converse, Inverse and Contrapositive

Important

High Frequency – 2025

For:

$$p \to q$$

10.1. Converse

$$\boxed{q \to p}$$

10.2. Inverse

$$\boxed{\neg p \to \neg q}$$

10.3. Contrapositive

$$\boxed{\neg q \to \neg p}$$

Memory Tip

Starting from $p \to q$:

- Converse $\to$ Swap
- Inverse $\to$ Negate
- Contrapositive $\to$ Swap + Negate

Important relationship:

$$\boxed{p \to q \equiv \neg q \to \neg p}$$

Therefore, a statement and its contrapositive are logically equivalent.

PYQ – 2025

The paper asked students to write the converse, inverse and contrapositive of an implication involving numerical statements and determine their truth values.

Previous Year Question

Exam Method

1. Identify p.
2. Identify q.
3. Write $p \to q$.
4. Converse: $q \to p$.
5. Inverse: $\neg p \to \neg q$.
6. Contrapositive: $\neg q \to \neg p$.
7. Evaluate their truth values.

11. Duality Principle

Important

Medium – 2025

To obtain the dual:

- interchange $\wedge$ and $\vee$;
- interchange T and F.

For set theory:

- interchange $\cup$ and $\cap$;
- interchange U and $\varnothing$.

Example:

$$p \wedge q$$

Dual:

$$\boxed{p \vee q}$$

12. Predicate Logic

Important

Syllabus topic – not directly seen in the 2024/2025 PYQs

A proposition has a definite truth value.
A **predicate** contains a variable.
Example:

$$P(x) : x > 5$$

This is not true or false until x is specified.
If $x = 7$:

$$P(7) : 7 > 5$$

which is True.

13. Well-Formed Formula (WFF)

A WFF is a logically correctly constructed expression.
Examples:

$$p \wedge q$$

$$\neg p$$

$$(p \vee q) \rightarrow r$$

Not a WFF:

$$\wedge pq$$

14. Quantifiers

14.1. Universal Quantifier

$$\boxed{\forall}$$

means "for all".
Example:

$$\forall x \in R, \quad x^2 \geq 0$$

14.2. Existential Quantifier

$$\boxed{\exists}$$

means "there exists at least one".
Example:

$$\exists x \in N, \quad x > 10$$

Memory Tip

$$\boxed{\forall = \text{ALL}}$$

$$\boxed{\exists = \text{AT LEAST ONE}}$$

15. Rules of Inference

Important

Medium – 2025

The 2025 paper specifically asked students to list any seven rules of inference with their logical implication and name.

15.1. 1. Modus Ponens

$$p \to q$$

$$p$$

Therefore:

$$\boxed{q}$$

15.2. 2. Modus Tollens

$$p \to q$$

$$\neg q$$

Therefore:

$$\boxed{\neg p}$$

15.3. 3. Hypothetical Syllogism

$$p \to q$$

$$q \to r$$

Therefore:

$$\boxed{p \to r}$$

15.4. 4. Disjunctive Syllogism

$$p \vee q$$

$$\neg p$$

Therefore:

$$\boxed{q}$$

15.5. 5. Addition

$$p$$

Therefore:

$$\boxed{p \vee q}$$

15.6. 6. Simplification

$$p \wedge q$$

Therefore:

$$\boxed{p}$$

15.7. 7. Conjunction

$$p$$

$$q$$

Therefore:

$$\boxed{p \wedge q}$$

16. Methods of Proof

Important

Medium – 2025

Important methods:

1. Direct proof
2. Proof by contrapositive
3. Proof by contradiction
4. Proof by cases

16.1. Direct Proof

Start with the given condition and logically reach the conclusion.

16.2. Contrapositive

To prove:

$$p \to q$$

prove:

$$\boxed{\neg q \to \neg p}$$

16.3. Contradiction

Assume the conclusion is false and obtain a contradiction.

17. Resolution Principle

Important

Syllabus topic – not directly seen in the two PYQs

Basic resolution rule:

$$(A \vee B), (\neg B \vee C)$$

gives:

$$\boxed{A \vee C}$$

The complementary literals B and $\neg B$ are eliminated.

MODULE 2 – SETS, RELATIONS AND FUNCTIONS

The current syllabus includes set representation, types, operations, algebra, inclusion-exclusion, relations, relation representation by matrix and graph, properties, equivalence relations and classes, and functions including mappings, injection, surjection, composition and inverse.

18. Set

Important

High Frequency – 2024 & 2025

Definition

A **set** is a well-defined collection of distinct objects.

Example:

$$A = \{1, 2, 3, 4\}$$

19. Types of Sets

19.1. Empty Set

$$\boxed{\varnothing}$$

contains no elements.

19.2. Singleton Set

Contains exactly one element.

$$A = \{5\}$$

19.3. Finite Set

Has a limited number of elements.

$$A = \{1, 2, 3\}$$

19.4. Infinite Set

Example:

$$N = \{1, 2, 3, \ldots\}$$

19.5. Equal Sets

Two sets have exactly the same elements.

19.6. Subset

$$A \subseteq B$$

means every element of A belongs to B.

20. Number of Subsets

Important

High Frequency – 2024 & 2025

If a set contains n elements:

$$\boxed{\text{Number of subsets} = 2^n}$$

If:

$$A = \{1, 2, 3\}$$

then:

$$2^3 = 8$$

subsets.
Number of proper subsets:

$$\boxed{2^n - 1}$$

PYQ – 2024

If A has n elements, the number of subsets of A is:

$$\boxed{2^n}$$

The subsets of:

$$A = \{1, 2, 3\}$$

are:

$$\boxed{\varnothing, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}}$$

21. Set Operations

Important

High Frequency – 2024

21.1. Union

$$\boxed{A \cup B}$$

Elements belonging to A, B, or both.

Memory Tip

$\cup =$ OR

21.2. Intersection

$$\boxed{A \cap B}$$

Elements common to both.

Memory Tip

$\cap =$ AND / COMMON

21.3. Difference

$$\boxed{A - B}$$

Elements in A but not in B.

$$A - B = A \cap B'$$

21.4. Complement

$$\boxed{A'}$$

Elements of the universal set that are not in A.

22. Cartesian Product

Important

High Frequency – 2024

For sets A and B:

$$\boxed{A \times B = \{(a, b) : a \in A,\ b \in B\}}$$

If:

$$|A| = m, \qquad |B| = n$$

then:

$$\boxed{|A \times B| = mn}$$

PYQ – 2024

If:

$$|A| = m, \qquad |B| = n$$

then:

$$\boxed{n(A \times B) = mn}$$

23. Algebra of Sets

Important laws:

23.1. Commutative Laws

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

23.2. Associative Laws

$$(A \cup B) \cup C = A \cup (B \cup C)$$

23.3. Distributive Law

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

23.4. Identity Laws

$$A \cup \varnothing = A$$

$$A \cap U = A$$

23.5. Complement Laws

$$A \cup A' = U$$

$$A \cap A' = \varnothing$$

23.6. De Morgan's Laws

$$\boxed{(A \cup B)' = A' \cap B'}$$

$$\boxed{(A \cap B)' = A' \cup B'}$$

24. Inclusion–Exclusion

Important

High Frequency – 2025

For two sets:

$$\boxed{n(A \cup B) = n(A) + n(B) - n(A \cap B)}$$

For three sets:

$$\boxed{\begin{aligned} n(A \cup B \cup C) = {} & n(A) + n(B) + n(C) \\ & - n(A \cap B) - n(B \cap C) - n(A \cap C) \\ & + n(A \cap B \cap C) \end{aligned}}$$

Memory Tip

$$\boxed{+ \text{ singles } - \text{ pairs } + \text{ triple}}$$

PYQ – 2025

49 study Physics, 37 study English and 21 study Biology.

Also:

$$n(P \cap E) = 9$$

$$n(E \cap B) = 5$$

$$n(P \cap B) = 4$$

and:

$$n(P \cap E \cap B) = 3.$$

Find the number of students.

$$\begin{aligned} n(P \cup E \cup B) &= 49 + 37 + 21 - 9 - 5 - 4 + 3 \\ &= 107 - 18 + 3 \\ &= \boxed{92} \end{aligned}$$

25. Ordered Pair

Important

High Frequency – 2025

An ordered pair is written:

$$(a, b)$$

Order matters.

$$(1, 2) \neq (2, 1)$$

Two ordered pairs are equal if:

$$(a, b) = (c, d)$$

then:

$$a = c, \qquad b = d.$$

PYQ – 2025

If:

$$(x^2 - 1, y + 3) = (3, 4)$$

find x and y.
Compare corresponding elements:

$$x^2 - 1 = 3$$

$$x^2 = 4$$

$$x = \pm 2$$

and:

$$y + 3 = 4$$

$$y = 1.$$

Therefore:

$$\boxed{x = \pm 2, \qquad y = 1}$$

26. Relation

Important

High Frequency – 2025

A relation R from A to B is a subset of:

$$A \times B.$$

Therefore:

$$\boxed{R \subseteq A \times B}$$

If $A = B$, then R is a relation **on** A.

27. Matrix Representation of a Relation

Suppose:

$$A = \{1, 2, 3\}$$

and R is a relation on A.

- Rows $\rightarrow$ first element
- Columns $\rightarrow$ second element
- Put 1 if the ordered pair belongs to R
- Put 0 otherwise

Example:

$$R = \{(1, 1), (1, 2), (2, 3), (3, 1)\}$$

Matrix:

$$\boxed{M_R = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}}$$

Memory Tip

Pair present = 1; pair absent = 0.

28. Graph Representation of Relation

Represent each element by a vertex.

For:

$$(a, b) \in R$$

draw an arrow:

$$a \to b$$

If:

$$(a, a) \in R$$

draw a loop at a.

PYQ – 2025

Let:

$$X = \{1, 2, 3, 4\}$$

and:

$$R = \{(x, y) : x > y\}.$$

The pairs satisfying $x > y$ are:

$$(2, 1)$$

$$(3, 1), (3, 2)$$

$$(4, 1), (4, 2), (4, 3).$$

Therefore:

$$R = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}.$$

The matrix, using the order $1, 2, 3, 4$, is:

$$\boxed{\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix}}$$

For the graph, draw:

$$2 \to 1$$

$$3 \to 1, \qquad 3 \to 2$$

$$4 \to 1, \qquad 4 \to 2, \qquad 4 \to 3.$$

29. Types of Relations

Important

High Frequency – 2025

29.1. Empty Relation

$$\boxed{R = \varnothing}$$

No ordered pair is present.

29.2. Universal Relation

$$\boxed{R = A \times A}$$

Every possible ordered pair is present.

29.3. Identity Relation

$$\boxed{I_A = \{(a, a) : a \in A\}}$$

Only self-pairs are present.

30. Reflexive Relation

A relation R on A is reflexive if:

$$\boxed{(a, a) \in R \quad \forall a \in A}$$

For:

$$A = \{1, 2, 3\}$$

we need:

$$(1, 1), (2, 2), (3, 3)$$

all to be present.
Extra pairs are allowed.
For example:

$$R = \{(1, 1), (2, 2), (3, 3), (1, 2)\}$$

is reflexive.

31. Irreflexive Relation

A relation is irreflexive if:

$$\boxed{(a,a) \notin R \quad \forall a \in A}$$

Therefore no self-pair is allowed.
Example:

$$R = \{(1,2),(2,3),(3,1)\}$$

on:

$$A = \{1,2,3\}$$

is irreflexive.

Memory Tip

Reflexive → ALL self-pairs
Irreflexive → NO self-pairs

32. Symmetric Relation

A relation is symmetric if:

$$\boxed{(a,b) \in R \Rightarrow (b,a) \in R}$$

Example:

$$R = \{(1,2),(2,1),(2,3),(3,2)\}$$

is symmetric.

33. Antisymmetric Relation

A relation is antisymmetric if:

$$\boxed{(a,b),(b,a) \in R \Rightarrow a = b}$$

In simple words:

Two different elements cannot be related in both directions.

For example:

$$(1,2) \in R$$

and:

$$(2,1) \notin R.$$

This is allowed.
But if both occur:

$$(1,2),(2,1) \in R$$

then antisymmetry is violated because:

$$1 \neq 2.$$

34. Transitive Relation

A relation is transitive if:

$$\boxed{(a,b) \in R,\ (b,c) \in R \Rightarrow (a,c) \in R}$$

Memory Tip

Go two steps → shortcut must exist.

If:

$$1 \to 2$$

and:

$$2 \to 3$$

then transitivity requires:

$$1 \to 3.$$

35. Quick Property Table

Property	Condition
Reflexive	All (a,a) present
Irreflexive	No (a,a) present
Symmetric	$a \to b \Rightarrow b \to a$
Antisymmetric	Both directions only if $a = b$
Transitive	$a \to b,\ b \to c \Rightarrow a \to c$

36. Equivalence Relation

Important

High Frequency – 2025

A relation is an **equivalence relation** if it is:

1. Reflexive
2. Symmetric
3. Transitive

Therefore:

$$\boxed{\text{Equivalence} = \text{Reflexive} + \text{Symmetric} + \text{Transitive}}$$

Memory Tip

$$\boxed{RST}$$

- R – Reflexive
- S – Symmetric
- T – Transitive

PYQ – 2025

If:

$$A = \{1, 2, 3, 4\},$$

give an example of a relation on A which is reflexive and transitive but not symmetric. A clean answer is the relation:

$$a \leq b.$$

Thus:

$$R = \{(a, b) \in A \times A : a \leq b\}.$$

Therefore:

$$\begin{aligned} R = \{&(1,1), (2,2), (3,3), (4,4), \\ &(1,2), (1,3), (1,4), \\ &(2,3), (2,4), (3,4)\}. \end{aligned}$$

Reflexive

All self-pairs occur.

Transitive

If:

$$a \leq b, \qquad b \leq c$$

then:

$$a \leq c.$$

Not Symmetric

$$1 \leq 2$$

but:

$$2 \not\leq 1.$$

Hence:

$$\boxed{R \text{ is reflexive and transitive but not symmetric.}}$$

37. Equivalence Class

If R is an equivalence relation on A, the equivalence class of a is:

$$\boxed{[a] = \{x \in A : xRa\}}$$

In simple words:

The equivalence class of a contains all elements related to a.

Example:

$$A = \{1, 2, 3, 4\}$$

Suppose the equivalence relation creates the groups:

$$\{1, 3\}, \qquad \{2, 4\}.$$

Then:

$$[1] = [3] = \{1, 3\}$$

and:

$$[2] = [4] = \{2, 4\}.$$

Memory Tip

Equivalence classes are groups of elements that are equivalent to each other.

38. Function

Important

High Frequency – 2024 & 2025

A function from A to B is written:

$$\boxed{f : A \to B}$$

Every element of A must have:

exactly one image in B.

39. Domain, Codomain and Range

For:

$$f : A \to B$$

- A = Domain
- B = Codomain
- Actual outputs = Range

Memory Tip

Domain = input
Range = actual output
Codomain = allowed output set

PYQ – 2024

If $f : A \to B$ is a function from A to B, then A is called:

$$\boxed{\text{Domain}}$$

40. Injective Function

Important

High Frequency – 2024 & 2025

A function is **injective (one-one)** if different inputs have different outputs.
Mathematically:

$$\boxed{f(a) = f(b) \Rightarrow a = b}$$

Simple idea:

No two different inputs share the same output.

PYQ – 2024

Given:

$$f(x) = 3x + 7$$

prove that f is one-one.
Let:

$$f(a) = f(b).$$

Then:

$$3a + 7 = 3b + 7$$

$$3a = 3b$$

$$a = b.$$

Hence:

$$\boxed{f \text{ is one-one.}}$$

41. Surjective Function

A function is **onto/surjective** if every element of the codomain has at least one preimage.
Therefore:

$$\boxed{\text{Range} = \text{Codomain}}$$

42. Bijective Function

A function is bijective if it is both:

- injective
- surjective

Therefore:

$$\boxed{\text{Bijective} = \text{One-one} + \text{Onto}}$$

43. Composition of Functions

Important

High Frequency – 2025

If:

$$f : A \to B$$

and:

$$g : B \to C$$

then:

$$\boxed{(g \circ f)(x) = g(f(x))}$$

Memory Tip

Right function first.
In:

$$g \circ f$$

do f first, then g.

44. Inverse Function

Important

High Frequency – 2025

The inverse function reverses the mapping.
If:

$$f : A \to B$$

then:

$$f^{-1} : B \to A.$$

A function has an inverse function when it is **bijective**.

PYQ – 2025

Find the inverse of:

$$f(x) = 6x + 3$$

and verify using composition.

Start with:

$$y = 6x + 3.$$

Interchange x and y:

$$x = 6y + 3.$$

Therefore:

$$x - 3 = 6y$$

and:

$$\boxed{f^{-1}(x) = \frac{x-3}{6}}$$

Verification

$$f(f^{-1}(x)) = 6\left(\frac{x-3}{6}\right) + 3$$

$$= x - 3 + 3$$

$$= \boxed{x}.$$

Similarly:

$$f^{-1}(f(x)) = \frac{(6x+3)-3}{6} = x.$$

Therefore the inverse is verified.

MODULE 3 – COUNTING TECHNIQUES

The syllabus covers basic counting principle, sum and product rules, permutations, combinations, pigeonhole principle, binomial theorem, recurrence relations and generating functions.

45. Rule of Sum

Important

Medium – Syllabus

If one task can be performed in m ways and another mutually exclusive task in n ways, then the total number of ways is:

$$\boxed{m+n}$$

Memory Tip

$$\boxed{\text{OR} \rightarrow \text{ADD}}$$

46. Rule of Product

If a process consists of two stages:

- first stage: m choices
- second stage: n choices

then the total number of ways is:

$$\boxed{mn}$$

Memory Tip

$$\boxed{\text{AND / sequence} \rightarrow \text{MULTIPLY}}$$

47. Permutations

Important

High Frequency – 2024 & 2025

Permutation means **arrangement where order matters**.

$$\boxed{{}^{n}P_r = \frac{n!}{(n-r)!}}$$

If all n objects are arranged:

$$\boxed{n!}$$

47.1. Repeated Objects

If there are n objects with repetitions:

$$\boxed{\frac{n!}{p!q!r!}}$$

PYQ – 2024

Find the number of permutations of the letters of **INDEPENDENCE**.

The word has 12 letters.

Repeated letters:

- $E = 4$
- $N = 3$
- $D = 2$

Therefore:

$$\frac{12!}{4!3!2!}$$

$$\boxed{16,632,000}$$

48. Combinations

Important

High Frequency – 2025

Combination means **selection where order does not matter**.

$$\boxed{{}^{n}C_r = \frac{n!}{r!(n-r)!}}$$

Memory Tip

$$\boxed{P = \text{Position}}$$

$$\boxed{C = \text{Choose}}$$

Permutation → Arrangement.
Combination → Selection.

49. Pigeonhole Principle

Important

Medium – 2025

If more than n objects are placed into n boxes, at least one box contains more than one object.

Mathematically:

$$\boxed{n+1 \text{ objects in } n \text{ boxes} \Rightarrow \text{one box has at least } 2}$$

PYQ – 2025

State Pigeonhole Principle.

Answer:

If $n+1$ objects are placed into n boxes, then at least one box contains at least two objects.

50. Binomial Theorem

Important

Medium – 2025

The binomial theorem is:

$$\boxed{(x+y)^n = \sum_{r=0}^{n} \binom{n}{r} x^{n-r} y^r}$$

General term:

$$\boxed{T_{r+1} = \binom{n}{r} x^{n-r} y^r}$$

50.1. Important: Term Number vs. r

If asked for the 10th term:

$$r + 1 = 10$$

Therefore:

$$\boxed{r = 9}$$

Then substitute $r = 9$ into the general term.

51. Middle Term

For:

$$(x + y)^n$$

there are:

$$n + 1$$

terms.

51.1. If n is Even

There is one middle term:

$$\boxed{T_{\frac{n}{2}+1}}$$

51.2. If n is Odd

There are two middle terms:

$$\boxed{T_{\frac{n+1}{2}}, \quad T_{\frac{n+3}{2}}}$$

PYQ – 2025

Identify the middle term in:

$$(x + 2y)^7.$$

Since $n = 7$, there are:

$$7 + 1 = 8$$

terms.
Therefore the middle terms are:

$$\boxed{T_4, \quad T_5}$$

52. Recurrence Relation

Important

Medium – 2025

A recurrence relation defines a sequence using previous terms.
Example:

$$a_n = a_{n-1} + a_{n-2}.$$

Initial conditions are needed to determine a specific sequence.

PYQ – 2025

The paper asked:

$$a_{n+2} = a_{n+1} + a_n$$

with:

$$a_0 = 0, \qquad a_1 = 1.$$

Then:

$$a_2 = a_1 + a_0 = 1$$

$$a_3 = a_2 + a_1 = 2$$

$$a_4 = 3$$

and so on.
Thus:

$$\boxed{0, 1, 1, 2, 3, 5, 8, \ldots}$$

which is the Fibonacci sequence.

53. Generating Functions

Important

Medium – 2025

For a sequence:

$$a_0, a_1, a_2, \ldots$$

the ordinary generating function is:

$$\boxed{G(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots}$$

Basic Idea

Convert a recurrence relation into an algebraic equation in x, solve for $G(x)$, and then obtain the sequence.

The 2025 paper specifically asked a recurrence to be solved by the method of generating functions.

MODULE 4 – ABSTRACT ALGEBRA AND LATTICES

The syllabus covers algebraic-system properties, homomorphism/isomorphism, groups, semigroups, subgroups, monoids, abelian groups and lattices.

54. Binary Operation

A binary operation $*$ on a set A is an operation that combines any two elements of A and gives an element of A.

$$\boxed{a * b \in A \quad \forall a, b \in A}$$

The result must remain inside the set.

55. Closure

A set is closed under an operation if applying the operation to any two elements of the set produces another element of the set.

Example:

$$a, b \in Z$$

then:

$$a + b \in Z.$$

Therefore integers are closed under addition.

56. Associative Property

$$\boxed{(a * b) * c = a * (b * c)}$$

The grouping does not affect the result.

57. Commutative Property

$$\boxed{a * b = b * a}$$

The order does not affect the result.

58. Semigroup

Important

High Frequency – 2024 & 2025

Definition

A **semigroup** is a non-empty set together with a binary operation that satisfies:

1. Closure
2. Associativity

Example:

$$\boxed{(N, +)}$$

is a semigroup.

PYQ – 2025

Define semigroup with example.

Exam Answer:
A semigroup is a non-empty set together with a binary operation that is closed and associative.
Example:

$$\boxed{(N, +)}$$

59. Monoid

Important

High Frequency – 2024 & 2025

A **monoid** is a semigroup with an identity element.
Therefore:

$$\boxed{\text{Monoid} = \text{Semigroup} + \text{Identity}}$$

Example:

$$(Z, +)$$

The identity is:

$$0$$

because:

$$a + 0 = 0 + a = a.$$

PYQ – 2025

Prove that:

$$(Z, +)$$

is a monoid.

Closure

For:

$$a, b \in Z$$

we have:

$$a + b \in Z.$$

Therefore addition is closed on Z.

Associativity

$$(a + b) + c = a + (b + c).$$

Therefore addition is associative.

Identity

$$a + 0 = 0 + a = a.$$

Therefore 0 is the identity.
Hence:

$$\boxed{(Z, +) \text{ is a monoid.}}$$

60. Group

A group is a non-empty set G with an operation satisfying:

1. Closure
2. Associativity
3. Identity
4. Inverse

Memory Tip

$$\boxed{C + A + I + I}$$

- Closure
- Associativity
- Identity
- Inverse

61. Abelian Group

A group is **abelian** if it is also commutative.

$$\boxed{a * b = b * a}$$

Therefore:

$$\boxed{\text{Abelian group} = \text{Group} + \text{Commutativity}}$$

62. Group Example: $(Z, +)$

Show:

$$(Z, +)$$

is a group.

Closure

$$a + b \in Z.$$

Associativity

$$(a + b) + c = a + (b + c).$$

Identity

The identity is:

$$0.$$

Inverse

For every $a \in Z$:

$$-a \in Z$$

and:

$$a + (-a) = 0.$$

Therefore:

$$\boxed{(Z, +) \text{ is a group.}}$$

Since:

$$a + b = b + a,$$

it is also an abelian group.

63. Subgroup

A non-empty subset H of a group G is a subgroup if H itself forms a group under the same operation.

Written:

$$\boxed{H \leq G}$$

PYQ – 2025

If H and K are subgroups of a group G, prove that:

$$H \cap K$$

is also a subgroup of G.

Proof

Since:

$$H, K \leq G,$$

take:

$$a, b \in H \cap K.$$

Then:

$$a, b \in H$$

and:

$$a, b \in K.$$

Since H is a subgroup:

$$ab^{-1} \in H.$$

Since K is a subgroup:

$$ab^{-1} \in K.$$

Therefore:

$$ab^{-1} \in H \cap K.$$

Hence, by the subgroup test:

$$\boxed{H \cap K \leq G}.$$

64. Homomorphism

Important

Medium – 2025

A function between algebraic structures is a homomorphism if it preserves the operation.

If:

$$f : G \to H$$

then:

$$\boxed{f(a * b) = f(a) \circ f(b)}$$

PYQ – 2025

The paper asked students to define a homomorphic function and show that:

$$g : \mathbb{Z} \to \mathbb{Z}_4$$

defined by:

$$g(a) = a \pmod 4$$

is a homomorphism.

To prove:

$$g(a + b) = g(a) + g(b) \pmod 4.$$

Since:

$$g(a + b) = (a + b) \pmod 4$$

and:

$$g(a) + g(b) = a \pmod 4 + b \pmod 4,$$

both are equal modulo 4.
Therefore:

$$\boxed{g \text{ is a homomorphism.}}$$

65. Isomorphism

An isomorphism is essentially a **bijective homomorphism**.
Therefore:

$$\boxed{\text{Isomorphism} = \text{Bijective Homomorphism}}$$

Memory Tip

Homo → preserves operation.
Iso → same structure in a one-to-one correspondence.

66. Lattice

Important

High Frequency – 2024 & 2025

A **lattice** is a partially ordered set in which every pair of elements has:

1. a greatest lower bound (GLB / meet)
2. a least upper bound (LUB / join)

For elements a, b:

$$\boxed{a \wedge b = \text{GLB}}$$

$$\boxed{a \vee b = \text{LUB}}$$

PYQ – 2025

Define a lattice.

Exam Answer:
A partially ordered set is called a lattice if every pair of elements has both a greatest lower bound and a least upper bound.

67. Hasse Diagram

A Hasse diagram is a simplified graphical representation of a partially ordered set.

Rules

- Do not draw loops.
- Do not draw arrows.
- Draw larger elements above smaller elements.
- Remove transitive edges.

68. Types of Lattices

Important

Medium – 2025

68.1. Bounded Lattice

Has:

- least element 0
- greatest element 1

68.2. Complemented Lattice

Every element has a complement.

68.3. Distributive Lattice

Satisfies:

$$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$$

and:

$$a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c).$$

68.4. Modular Lattice

Satisfies the modular law.

IMPORTANT EXAM DIFFERENCE TABLES

69. Permutation vs Combination

Permutation	Combination
Arrangement	**Selection**
Order matters	**Order does not matter**
${}^nP_r$	${}^nC_r$
$\frac{n!}{(n-r)!}$	$\frac{n!}{r!(n-r)!}$

Memory Tip

$\boxed{P = \text{Position}}$

$\boxed{C = \text{Choose}}$

70. Reflexive vs Irreflexive

Reflexive	Irreflexive
Every self-pair present	**No self-pair present**
$(a, a) \in R$	$(a, a) \notin R$
ALL self-pairs	**NO self-pairs**

71. Symmetric vs Antisymmetric

Symmetric	Antisymmetric
$aRb \Rightarrow bRa$	aRb **and** $bRa \Rightarrow a = b$
Both directions are required	**Both directions for different elements are forbidden**
$(1, 2)$ **requires** $(2, 1)$	$(1, 2)$ **and** $(2, 1)$ **cannot both occur**

Memory Tip

Symmetric: "Go there → come back."
Antisymmetric: "Different elements cannot go both ways."

72. Relation vs Function

Relation	Function
Any subset of $A \times B$	**Special type of relation**
An input may have multiple outputs	**Every input has exactly one output**
An input may have no output	**Every domain element must have an output**

73. Injective vs Surjective

Injective	Surjective
One-one	**Onto**
Different inputs $\rightarrow$ different outputs	**Every codomain element is reached**
No two inputs share an output	**Range = Codomain**

74. Semigroup vs Monoid vs Group

Structure	Required Properties
Semigroup	Closure + Associativity
Monoid	Semigroup + Identity
Group	Monoid + Inverse
Abelian Group	Group + Commutativity

Memory Tip

$$\boxed{\text{Semigroup} \rightarrow \text{Monoid} \rightarrow \text{Group} \rightarrow \text{Abelian Group}}$$

Think:

$$\boxed{C \rightarrow CA \rightarrow CAI \rightarrow CAII}$$

75. Homomorphism vs Isomorphism

Homomorphism	Isomorphism
Preserves operation	**Preserves operation**
Need not be bijective	**Must be bijective**
Structure-preserving mapping	**One-to-one structure-preserving mapping**

PYQ-BASED REVISION LIST

These are the concepts students should be able to solve without looking at their notes.

Logic

1. Construct truth tables.
2. Identify tautology, contradiction and contingency.
3. Prove logical equivalence.
4. Find converse, inverse and contrapositive.
5. Write dual.
6. Apply rules of inference.
7. Establish validity using a method of proof.

Sets

8. Find all subsets.
9. Find $A \cup B$, $A \cap B$.
10. Use Cartesian product.
11. Solve inclusion-exclusion problems.
12. Draw/set expressions using Venn diagrams.

Relations

13. Give examples of reflexive, irreflexive, symmetric, antisymmetric and transitive relations.
14. Check whether a relation is an equivalence relation.
15. Construct relation matrix.
16. Draw relation graph.
17. Find equivalence classes.

Functions

18. Prove a function is one-one.
19. Identify domain, codomain and range.
20. Find composition.
21. Find inverse and verify using composition.
22. Determine whether a function is onto.

Counting

23. Use sum/product rule.
24. Solve permutation problems.
25. Solve combination problems.
26. Apply pigeonhole principle.
27. Find general/middle terms in binomial expansion.
28. Solve basic recurrence relations.
29. Understand generating functions.

Algebra

30. Verify closure, associativity, identity and inverse.
31. Prove a set is a semigroup/monoid/group.
32. Prove subgroup results.
33. Define/prove homomorphism.
34. Distinguish homomorphism and isomorphism.
35. Identify/check lattices.
36. Draw Hasse diagrams.

WHAT THE 2024 AND 2025 PAPERS TELL US

There is a useful pattern in the two papers.

2024 Strongly Tested

- Sets
- Set operations
- Cartesian product
- Functions
- Permutations
- Graph/tree topics
- Semigroup/monoid
- Groups
- Lattice/Hasse diagram

The 2024 paper also contains graph theory and tree questions, but those topics do not appear in the current 3341 syllabus. Therefore, those old graph/tree questions should not automatically be treated as core material for the current syllabus unless separately instructed.

2025 Strongly Tested

- Propositional logic
- Tautology
- Logical equivalence
- Relations
- Relation matrix/graph
- Sets/subsets
- Functions
- Inclusion-exclusion
- Inverse function
- Permutations/combinations
- Pigeonhole principle
- Binomial theorem
- Recurrence relations
- Generating functions
- Semigroup/monoid
- Subgroup
- Homomorphism
- Lattice

This makes the 2025 paper particularly useful for identifying the current syllabus emphasis.

FINAL ONE-PAGE MEMORY SHEET

LOGIC

$$\boxed{p \to q \equiv \neg p \vee q}$$

$$\boxed{\text{Tautology} = \text{always T}}$$

$$\boxed{\text{Contradiction} = \text{always F}}$$

$$\boxed{\text{Contingency} = \text{T and F}}$$

$$\boxed{\text{Contrapositive of } p \to q = \neg q \to \neg p}$$

$$\boxed{\text{Equivalence} = \text{same truth table}}$$

SETS

$$\boxed{\text{Subsets} = 2^n}$$

$$\boxed{|A \times B| = |A||B|}$$

$$\boxed{n(A \cup B) = n(A) + n(B) - n(A \cap B)}$$

$$\boxed{\text{Three sets} = +\text{singles} - \text{pairs} + \text{triple}}$$

RELATIONS

$$\boxed{\text{Reflexive} = \text{ALL self-pairs}}$$

$$\boxed{\text{Irreflexive} = \text{NO self-pairs}}$$

$$\boxed{\text{Symmetric: } aRb \Rightarrow bRa}$$

$$\boxed{\text{Antisymmetric: } aRb, bRa \Rightarrow a = b}$$

$$\boxed{\text{Transitive: } aRb, bRc \Rightarrow aRc}$$

$$\boxed{\text{Equivalence} = \text{R+S+T}}$$

FUNCTIONS

$$\boxed{\text{Injective} = \text{one-one}}$$

$$\boxed{\text{Surjective} = \text{onto}}$$

$$\boxed{\text{Bijective} = \text{one-one} + \text{onto}}$$

$$\boxed{(g \circ f)(x) = g(f(x))}$$

$$\boxed{f^{-1}(f(x)) = x}$$

COUNTING

$$\boxed{\text{OR} \rightarrow \text{ADD}}$$

$$\boxed{\text{AND} \rightarrow \text{MULTIPLY}}$$

$$\boxed{{}^{n}P_r = \frac{n!}{(n-r)!}}$$

$$\boxed{{}^{n}C_r = \frac{n!}{r!(n-r)!}}$$

$$\boxed{P = \text{Position}, \qquad C = \text{Choose}}$$

ALGEBRA

$$\boxed{\text{Semigroup} = C + A}$$

$$\boxed{\text{Monoid} = C + A + I}$$

$$\boxed{\text{Group} = C + A + I + \text{Inverse}}$$

$$\boxed{\text{Abelian Group} = \text{Group} + \text{Commutative}}$$

$$\boxed{\text{Isomorphism} = \text{Bijective Homomorphism}}$$

LATTICE

Lattice = LUB + GLB for every pair

Artificial Intelligence and Machine Learning